



Day 5 — Incremental Update & Demo (15–30 minutes)

1. Incremental Fine-Tuning Script

Create `src/incremental_tune.py`:

```
import argparse
import json
import os

def load_detector(path="model/final/detector.json"):
    with open(path) as f:
        return json.load(f)

def save_detector(data, path="model/final/detector.json"):
    os.makedirs(os.path.dirname(path), exist_ok=True)
    with open(path, "w") as f:
        json.dump(data, f, indent=2)

def incremental_update(new_logs, epochs=1):
    data = load_detector()
    patterns = set(data["normal_patterns"])
    # Read new benign sequences
    with open(new_logs) as f:
        for _ in range(epochs):
            for line in f:
                seq = line.strip()
                patterns.add(seq)
    data["normal_patterns"] = list(patterns)
    save_detector(data)
    print(f"✓ Incremental update added {len(patterns)} total patterns")

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--new_logs", required=True, help="New benign log file")
    parser.add_argument("--epochs", type=int, default=1)
    args = parser.parse_args()
    incremental_update(args.new_logs, args.epochs)
```

Usage (1 minute):

```
# Prepare new benign data
cp data/normalized.txt data/new_benign.txt # or your new logs
# Incremental update
```

```
python src/incremental_tune.py --new_logs data/new_benign.txt --epochs 1
# Restart sidecar
pkill -f "uvicorn src.server" && uvicorn src.server:app --reload &
```

2. Demo Attack Script

Save as demo/attacks.sh:

```
#!/usr/bin/env bash
set -e

echo "=== BENIGN REQUEST ==="
curl -s -o /dev/null -w "Status: %{http_code}\n" http://localhost/ -H "User-Agent: curl"

echo -e "\n=== SQLi ATTEMPT ==="
curl -s -o /dev/null -w "Status: %{http_code}\n" \
    "http://localhost/api/getItems?id=1 UNION SELECT password FROM users" -H "User-Agent: c

echo -e "\n=== XSS ATTEMPT ==="
curl -s -o /dev/null -w "Status: %{http_code}\n" \
    "http://localhost/search?q=<script>alert(1)</script>"
```

Make executable:

```
chmod +x demo/attacks.sh
```

3. Demo Sequence (5 minutes)

1. **Start Tomcat** (upstream at :8081)
2. **Start Sidecar:**

```
uvicorn src.server:app --reload --port 8080 &
```

3. **Start OpenResty:**

```
openresty -s reload
```

4. **Generate Benign Traffic:**

```
python scripts/gen_logs.py
while read ln; do curl -s -o /dev/null "http://localhost${ln#* }"; done < data/benign
```

5. **Monitor Detections** (terminal 1):

```
tail -f /var/log/waf_detections.log
```

6. **Run Attacks** (terminal 2):

```
demo/attacks.sh  
tail -n 10 /var/log/waf_detections.log
```

4. Evaluation Metrics

- **True Positives:** Ensure attack payloads from `attacks.sh` appear in `waf_detections.log`.
- **False Positives:** After 2000 benign requests, run:

```
wc -l /var/log/waf_detections.log
```

Compute $FP_rate = \text{detections} / 2000 * 100\%$ (target < 2%).

- **Latency:** Measure with and without sidecar:

```
# Without sidecar (direct upstream)  
ab -n1000 -c10 http://127.0.0.1/  
# With sidecar+OpenResty  
ab -n1000 -c10 http://127.0.0.1/
```

Compare `Requests per second` – should be nearly identical.

Total setup time: ~15–30 minutes for incremental update, demo scripts, and evaluation-ready workflow.